

Modeling and Analysis of Dynamic Systems: Vibration of Mechanical Systems

Overview and Background

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Course Syllabus

Term: 2026 Summer Term (2 weeks course, 5 – 14 August)
Class Schedule: 18:50 – 21:15 (with two 5-min breaks)
Instructor: Professor Chin An Tan, Ph.D., Fellow ASME
Dept. of Mechanical Engineering, Wayne State University, Detroit, MI 48202, USA
Contacts: email (tan@wayne.edu), DingTalk or WeChat
Office Hours: About 30 minutes before class in the classroom or by emails/texts

Course materials will be based on lecture notes developed by the instructor. The following is list of reference textbooks for further reading. Students are strongly encouraged to discuss with the instructor on any course-related matters.

Reference textbooks:

-  Rao, S. S., *Mechanical Vibrations*, 2018, 6th Edition, Pearson. ISBN-13: 978-1-292-17860-8.
-  Inman, D. J., *Engineering Vibration*, Prentice Hall, Fourth edition, 2014, ISBN-13: 978-0132281737.
-  Meirovitch, L., *Fundamentals of Vibrations*, Waveland Press, 2001, ISBN 1-57766-691-7.
(textbook; focus more on the fundamental theory and mathematics)
-  Ginsberg, J. H., *Mechanical and Structural Vibrations: Theory and Applications*, John Wiley and Sons, 2001, ISBN 0-471-37084-3.
-  Palm, William J., III, *Introduction to MATLAB for Engineers*, McGraw-Hill, Third Edition, 2011, ISBN 0-07-353487-0. (Reference for computational problems)

Course Syllabus

Homework: Homework assignments will be assigned after each lecture or every other lecture, depending on the progress of the class. No late assignment will be accepted, except under unforeseen circumstances. **Submit your homework in class (in printed version) by the due date.**

Tests: Final exam, Few quizzes (roughly every 1 or 2 lectures)

Grading:	Homework	25%
	Project	20%
	Quizzes & in-class tests	25%
	Final Exam	25%
	Class attendance	5%
	Active classroom participation	bonus

Course final grades will be based on percentages scaled with respect to the highest total score or 950 points (maximum 1,000 points), whichever is higher.

Guidelines for assigning grades:

(A = 95%+, A- = 90%+, B+ = 87%+, B = 83%+, B- = 80%+, C+ = 77%+, C = 73%+, C- = 70%+, D+ = 65%+, D = 60%+, D- = 55%+, F = less than 55%)

Course Syllabus

Useful web links:

Modal Analysis Group on LinkedIn

Professor Dan Russell animation website <http://www.acs.psu.edu/drussell/demos.html>

LECTURE	OUTLINE OF TOPICS
<i>Foundation and Basic Concepts of Vibration of SDOF Systems</i>	
August 05 (Wednesday)	Course administration Introduction to vibration; Free vibration of single-degree-of-freedom undamped and damped systems; modeling and equations of motion; concept of natural frequency; response solutions; Measurement of damping; examples and MATLAB solutions
August 06 (Thursday)	Forced vibration of SDOF systems to harmonic excitations Frequency response solutions; concept of resonance, examples and MATLAB solutions
August 07 (Friday)	Vibration isolation techniques; Forced response of SDOF systems to arbitrary non-periodic excitations; convolution integral, Laplace transformation method, other solution methods; examples and MATLAB solutions
<i>Vibration of 2DOF Systems</i>	
August 10 (Monday)	Free vibration of two-degrees-of-freedom systems; modeling and equations of motion, Lagrange's equations Concepts of coupling, decoupled equations of motion and modal analysis
August 11 (Tuesday)	Forced vibration of 2DOF systems Frequency response, examples and MATLAB solutions

Course Syllabus (cont.)

Vibration of Multi-Degree-of-Freedom Systems

August 12 (Wednesday)	Modeling of MDOF systems; equations of motion; Lagrange's equations Orthogonality of vibration modes (modal vectors); modal analysis; necessary and sufficient conditions for modal decoupling and physical significance Examples and MATLAB solutions
August 13 (Thursday)	Free and forced responses of undamped systems Proportionally damped systems; examples and MATLAB solutions Discussion of Project
August 14 (Friday)	Free and forced responses of damped systems Discussion of Project Introduction to modal analysis (optional)
August 17 (Monday)	Final Examination
August 19 (Wednesday)	Project due at 23:00

Motivation for Studying this Subject

- *What is the subject of modeling and analysis of dynamic systems about? What do we study in the subject?*

Mechanical and electrical systems can be modeled and analysed using concepts of system dynamics. For example, smart material and structural systems (SMSS) can be studied using tools of system dynamics. Motions of dynamic systems normally result in either complex or periodic motions (vibrations). Vibrations are associated with frequencies and frequency domain analysis is very useful tool for analysis and design of dynamic systems, with applications to energy harvesting, structural vibration control, etc.

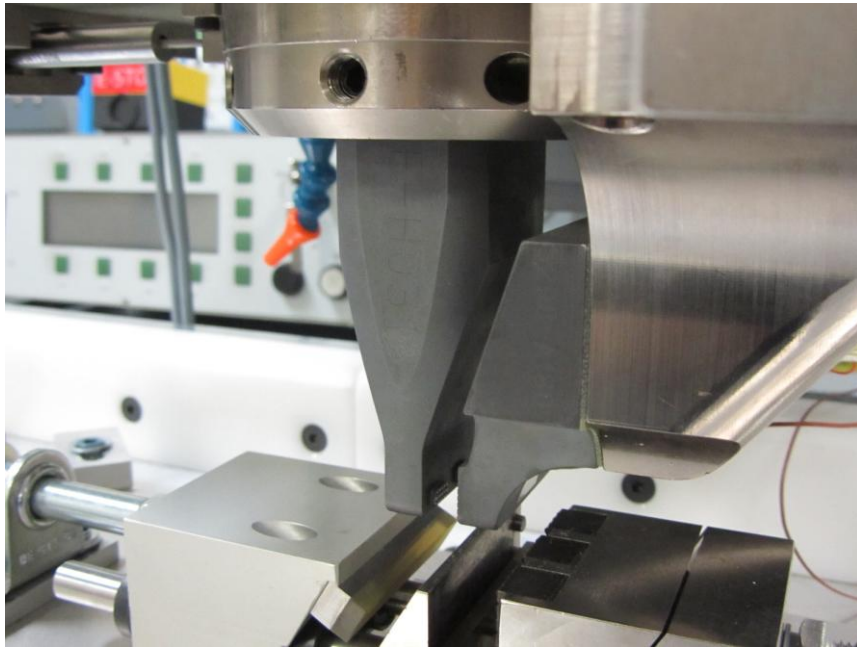
Motivation for Studying this Subject

- *Why is the subject of modeling and analysis of dynamic systems important?*

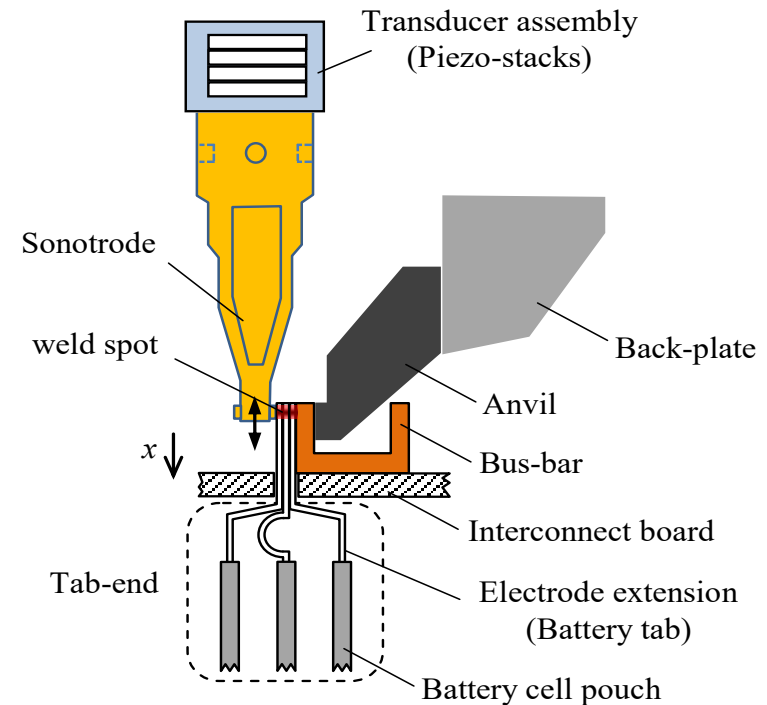
It provides a foundation for the response or vibration analysis of dynamic systems. Vibration occurs in a wide range of systems. While there are many beneficial applications of vibrations, however, unwanted/undesirable vibrations must be controlled, possibly by smart materials, to prevent structural and even catastrophic failures. In this course, we focus on the fundamentals of vibration of mechanical and structural systems.

Applications

- Useful applications of vibrations
 - ultrasonic vibration welding (frequency ≥ 20 kHz)
 - dental teeth cleaning, ...



Bond welds for automotive battery tabs



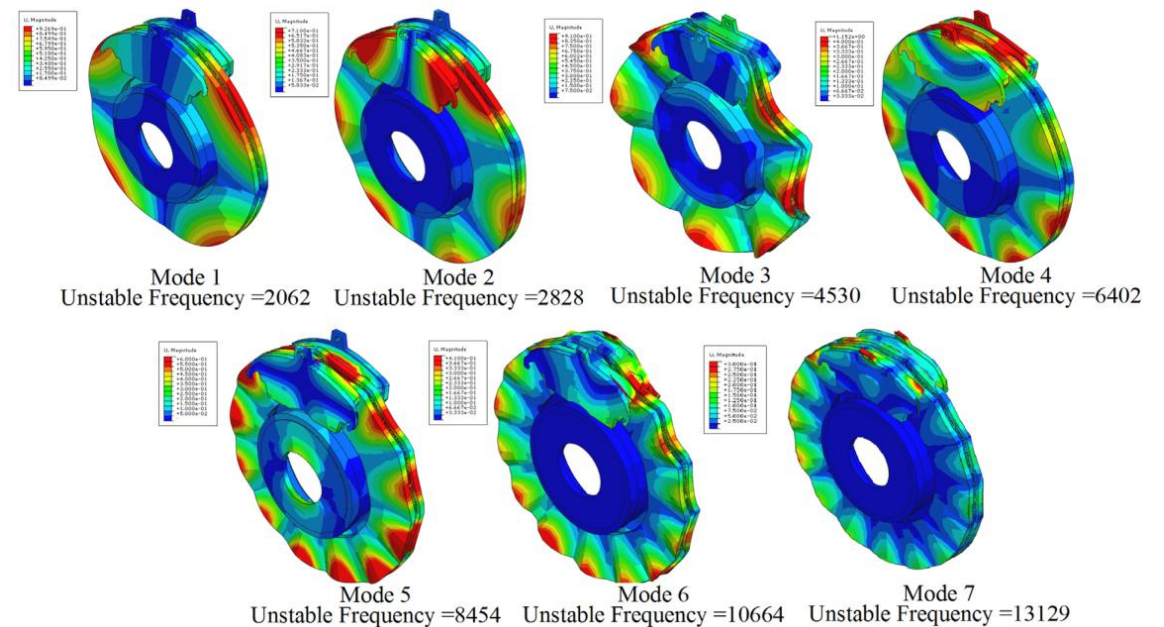
Kang, Cai, & Tan, *Fundamental Dynamics of Ultrasonic Welding*, in Monograph: Ultrasonic Welding of Lithium-Ion Batteries, 2017.

Applications

- Useful applications of vibrations
 - understand physics for design purpose
 - e.g., aircraft turbine blades, disc brakes, disc drives, ...



<https://www.youtube.com/watch?v=v4ELxKKT5Rw>



Karamoozian, Tan, & Wang, Squeal analysis of thin-walled lattice brake disc structure, *Materials and Design*, 2018.

Applications

- Useful applications of vibrations
 - breaking up and conditioning of concretes
 - e.g., jack hammer, tamping rammer



<https://www.youtube.com/watch?v=Cm3o8-G5luI>



<https://www.youtube.com/watch?v=SDnuDp4rh2g>

Applications

- Unwanted and undesirable vibrations
 - earthquakes causing collapse of structures



Bedford and Folwer, *Engineering Dynamics*, 5th edition

Applications

- Unwanted and undesirable vibrations
 - excessive vibration of bridges due to *design flaws*



The London Millennium Bridge
on Opening Day, 10 June 2000

Millennium Bridge: Incomplete Analysis

Something went wrong in the design analysis?

- The Millennium Bridge over the Thames in London was closed on its opening weekend in May 2000, since it wobbled more than expected.
- The simulations done during the design process handled the vertical force (which is all that is required by the British Standards Institution) of a pedestrian at around 2 Hz, but not the horizontal at about 1 Hz. What happened was that the slight wobbling (within tolerances) due to the wind caused the pedestrians to walk in step (synchronous walking) which made the bridge wobble even more.
- The bridge was reopened in 2002, after 37 viscous dampers and 54 tuned mass dampers were installed and all the modifications had been carefully tested. The modifications were completely successful.



Source: <http://www.nsc.liu.se/wg25/book/ch1/>

Applications

- Control of unwanted and undesirable vibrations
 - Control of Taipei 101 building under wind loading
 - Pendulum tuned mass damper

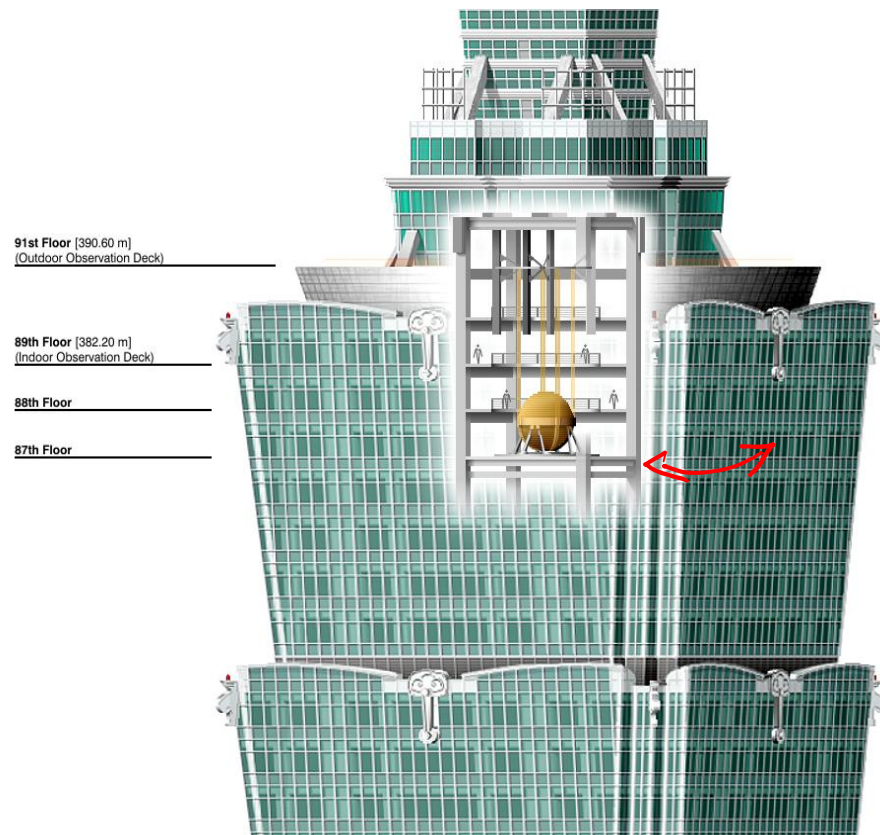
The *Taipei 101*, located in Taipei, Taiwan, was the tallest building in the world until the completion of the BurjKhalifa in Dubai in 2009.

It utilizes a 660-tonne steel *pendulum* located at the very top of the building. The steel pendulum acts as a tuned mass damping (TMD) mechanism to dissipate the vibration energies introduced by the environment.



Applications

- Control of unwanted and undesirable vibrations by *tuning*
 - Experimental simulation of the pendulum TMD

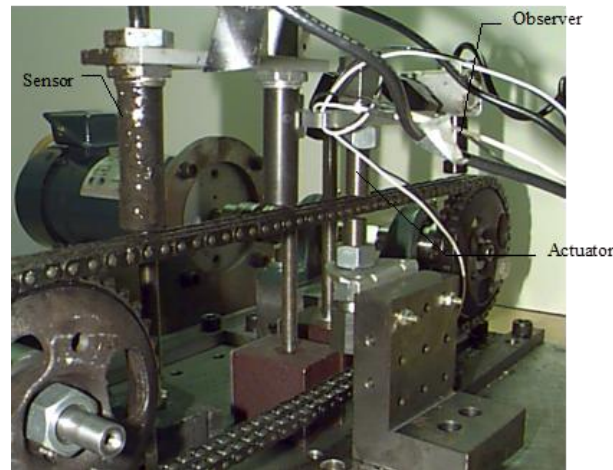
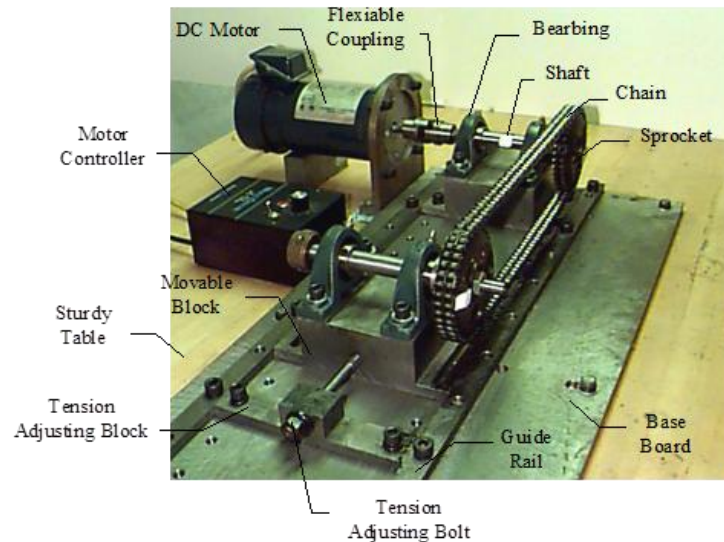


<https://www.youtube.com/watch?v=VCxm3vTWgvU>

Applications

- Active control of structures (via electrical inputs) by:
 - control algorithms via actuation
 - smart multifunctional materials

Experimental study of active vibration control of automotive timing chains.



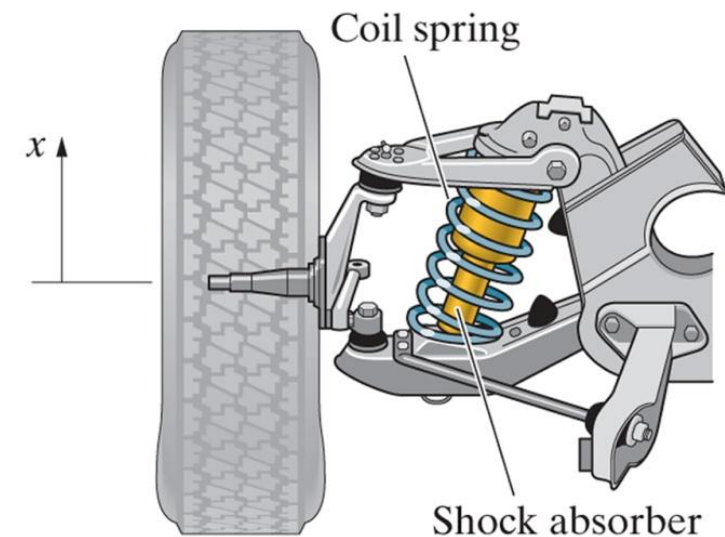
Critical Thinking and Key Observations

- Vibration responses are characterized by amplitudes and frequencies
- Vibrations are excited by external disturbances
 - initial and usually transient disturbances such as wind loads; this is called free vibration
 - persistent disturbances such as machinery loads; this is called forced vibration

Critical Thinking and Key Observations

- There are three basic structural elements involved in vibration
 - inertia or mass
 - stiffness or structural flexibility
 - damping or energy dissipation

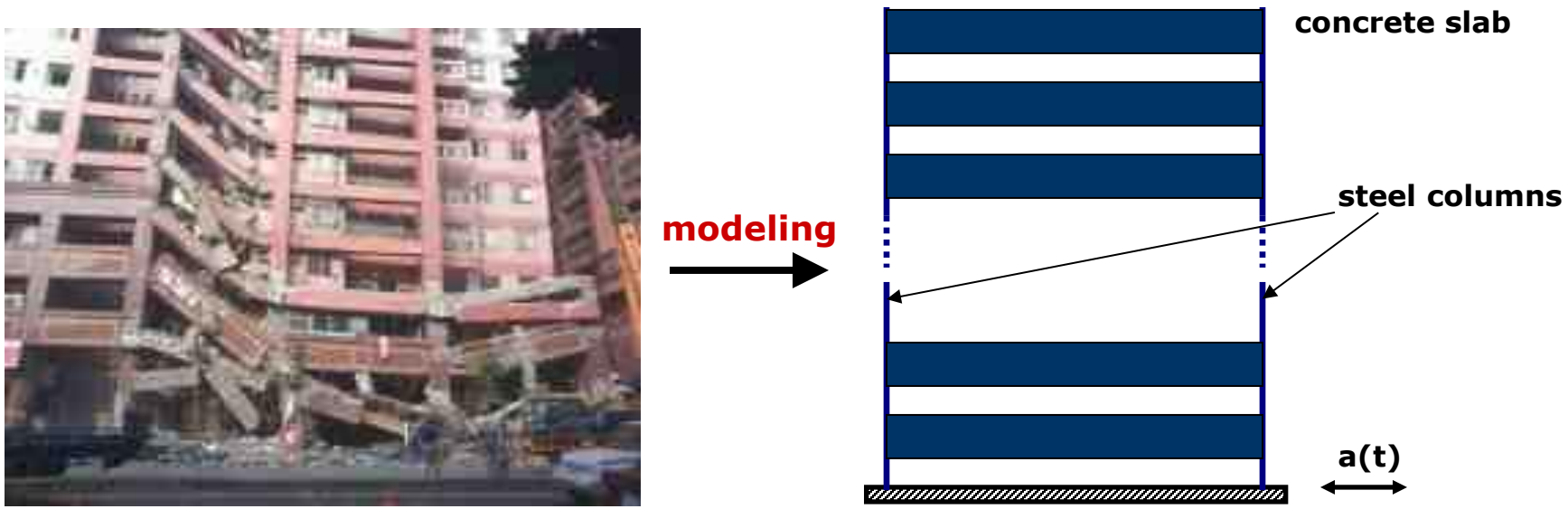
An automotive suspension system is designed to allow vibration via the coil spring and dissipate energy via the shock absorber. Proper designs require extensive study of the vibration characteristics and testing. Suspension design is critical in vehicle stability in autonomous driving.



Bedford and Fowler, *Engineering Dynamics*, 6th edition

Physical Modeling

- Different levels of modeling
 - Single-degree-of-freedom (SDOF) systems represent the simplest type of vibratory systems. They are approximations of real-world problems but often reveal the essential vibration characteristics.
 - Lumped multiple-degree-of-freedom (MDOF) systems
 - Distributed parameter (distributed mass) systems, e.g., modeling of smart materials using constitutive relations
- Linear vibration: small amplitude assumption
Nonlinear vibration: complicated phenomena such as chaos



Example: Pancake collapse of a building under earthquake excitation.

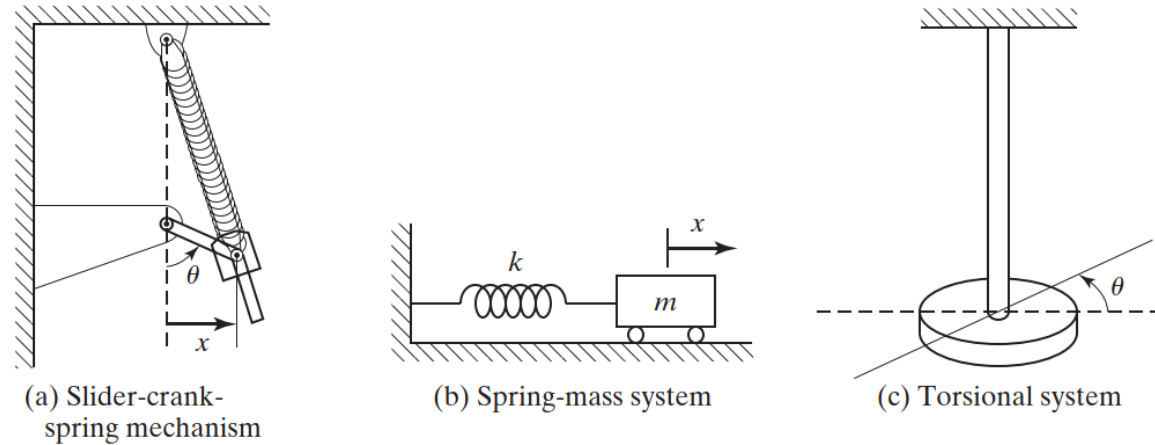
Procedures:

- Physical modeling and derivation of the governing equations of motion by Newton's Laws, Lagrange's equations, etc.
- Solution of the equations of motion by analytical, semi-analytical or numerical methods (computational tool such as MATLAB, Mathematica).
- Study of the qualitative and quantitative behaviour of the system response.
- Applications: frequency analysis, stability, parameters and damage identification, etc.

Number of Degrees of Freedom

Degree of freedom (DOF) is the minimum number of independent coordinates required to completely describe the motion.

■ Single-DOF



Rao, *Mechanical Vibrations*, 6th edition, 2018.

FIGURE 1.11 Single-degree-of-freedom systems.

■ 2-DOF

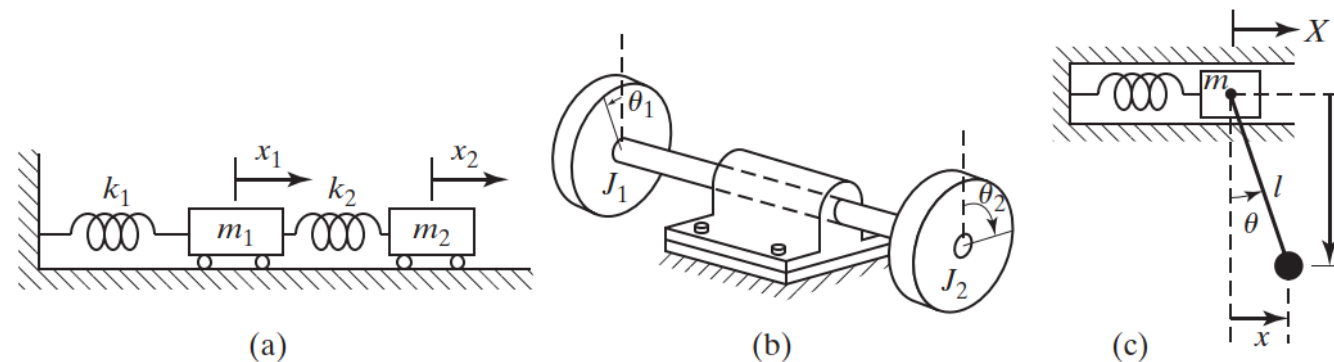
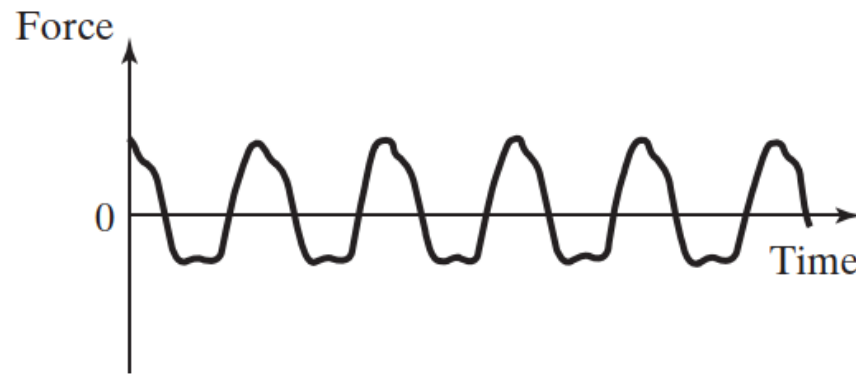


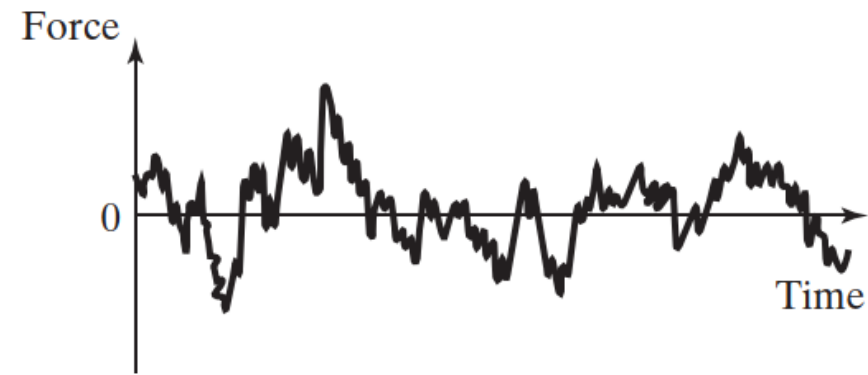
FIGURE 1.12 Two-degree-of-freedom systems.

Types of Excitations

- Deterministic and random excitations (in this class, we focus on deterministic; random requires knowledge of statistics for modeling)



(a) A deterministic (periodic) excitation



(b) A random excitation

Rao, *Mechanical Vibrations*,
6th edition, 2018.

FIGURE 1.15 Deterministic and random excitations.

General Procedure for Analysis

These are what we will cover in this course:

- Mathematical modeling (assumptions, choice of coordinates)
- Apply dynamics to derive governing equations of motion
- Apply concepts of differential equations to obtain analytical solutions (usually possible for linear systems)
- Understanding of fundamental concepts
- Numerical solutions
- Physical interpretation of results and critical thinking